

ASSCOIATION RULE MINING

DEFINTIONS



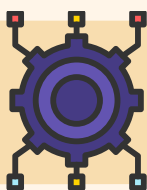
- Association rule mining is a data mining technique used to
- Discover interesting relations between variables or seemingly unrelated data in large datasets by implementing if-else statements.
 - Widely used in data mining to detect frequent patterns,correlations or associations.

PURPOSE



- Uncover hidden patterns in large datasets (e.g., market basket analysis)
- Help businesses understand customer behavior
- Predict behavior
- Make informed decisions in marketing,retail and health

APRIORI ALGORITHM



- Identifies frequent itemsets using a bottom-up approach
- Uses support threshold to eliminate infrequent items
- Generates strong association rules from frequent itemsets using confidence threshold

- Steps:
- Count individual items and eliminate below minimum support
 - Generate candidate itemsets (pairs, triples, etc.)
 - Repeat until no more frequent itemsets

REAL WORLD APPLICATIONS



Market Basket Analysis

- Retailers use Association Rules to identify items frequently purchased together,helping with:
- Store layout design
 - Product bundling
 - Targeted promotions
- Example:**
- Rule: If a customer buys Diapers → They also buy Baby wipes**
 - Marketing team can promote baby wipes discounts to diaper buyers.**

DATASET



Transaction ID	Item Bought
T1	Bread,Milk
T2	Bread,Milk,Beer,Eggs
T3	Bread,Diaper,Cola,Milk
T4	Bread,Milk,Diaper,Eggs
T5	Milk,Diaper,Beer,Cola

- ✓ **Frequent Itemsets (Support ≥ 60%):**
- {Bread,Milk} → Appears in 4/5 = 80%**
 - {Diaper,Milk} → Appears in 3/5 = 60%**
- 🔄 **Sample Association Rules:**
- Rule: {Bread} → {Milk}**
 - Support = 80%, Confidence = 100%, Lift = 1.0
 - Rule: {Milk} → {Bread}**
 - Support = 80%, Confidence = 80%, Lift = 1.0

KEY METRICS

Support: Measures how frequently the itemset appears in the dataset

Formula:



Support(A→B) =
Transactions containing both A & B

Total transactions

- Example: If 3 out of 5 transactions contain {Bread, Milk}, Support = 60%

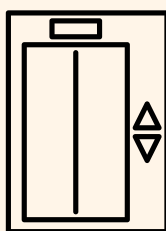
Confidence: Measures the likelihood of B given A

Formula:



Confidence(A→B) =
Support(A ∪ B)
Support(A)

Lift: Indicates how much more likely B is to occur with A than by itself.



Formula:

Lift(A→B) =
Confidence (A→B)
Support(A)

- Lift > 1: Positive correlation
- Lift < 1: Negative correlation
- Lift = 1: No correlation